Modern Techniques in The Modelling of High Voltage Power Systems

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***Abstract*— All components of the high voltage power systems (such as overhead power lines and substations) must meet power requirements and ensure large distance transmission and distribution of electrical energy to the end user. With the continuous demand of energy worldwide and the need to extend the power grid, new techniques are required to be developed in order to build or upgrade overhead power lines and substations efficiently, accurately and at lower costs. The purpose of this paper is to highlight the use of two modern design techniques in the modelling of high voltage power systems: PLS-CADD/TOWER and Revit; for technical parameters, environmental features and visualization tools. The PLS-CADD/TOWER software is used in the modelling and analysis of overhead power lines. Revit is used in the modelling of substations. In addition, both components can be integrated in a BIM (Building Information Modelling) environment. With the help of these tools all the information required for a project can be merged and used as design framework adaptable between projects.**

Keywords—overhead power lines & substations; BIM, Revit & PLS-CADD modelling & simulation software

I. Introduction

The power grid is one of the main topics of the major summits worldwide as the use of energy has been drastically increasing over the past 10 to 20 years. The demand for energy 24/7 has led to the need of expansion and upgrade of the power grid, to increase the level of voltage stability and to ensure continuity in supply.

At the same time, engineers must take into consideration the sustainability factor which can be interpreted as the balance between the following 3 pillars: environmental, social and economic impact.

The way to achieve sustainability is to use the modern techniques and the knowledge available nowadays. Over the years, technical progress can be seen not only in the process of production, transmission and distribution of electrical energy, but also in the way of modelling the overhead power lines and the power substations.

Power transmission utilities in the present days are more dependent on the standard designs for constructing transmission to minimize the capital cost. Economic studies indicate that overhead power lines employing optimized designs and scientific cost management methodologies for construction are significant in achieving much higher savings in capital investments [1].

Technology evolution continuously developed new ways to automatize the design process: automatic calculation of sag tension, automatic spotting of tower loads and 3D model of entire overhead power lines and substations; by also embracing the economic pillar. Such automatic design process allows an integrated check of the system and calculation of forces and moments for the design of the tower as well as the foundations. This is required for both line upgrading and new overhead power lines where updated wind mappings, new standards and new additional specifications for design must be considered. Integrating PLS-CADD and TOWER provides the assessment of structural analysis for these situations [2].

The main purpose of this paper is to make use of the available tools, such as Building Information Modelling (BIM) and 3D modelling software like PLS-CADD and TOWER, which adds value to the final solution by reducing delivery time and improving the level of accuracy due to the setup of a design framework. Working with these powerful tools and the results thus driven has emphasized new possibilities for an enhanced design and modelling quality that covers both technical and environmental aspects by merging solutions from Revit and PLS-CADD.

II. Design Considerations

The design and modelling of overhead power lines and substations includes several stages. The use of modern tools techniques together with BIM environment allows designers to create, from the early phases of the project, a reusable database built on various technical and environmental criteria (terrain profile, weather and clearances conditions) and to integrate the information in the design framework. One of the important features of tools like PLS-CADD/TOWER is the possibility to build, compare and simulate alternatives starting from the input data and the adaptability of the model to the specific requirements of the customer [3]. The use of BIM modelling shows great potential: the model is the document manager for the collection of the whole

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	Feat. Code	Feature Description	Prof Symbol	Plan Symbol	Line From Feature Top To Bottom	Aerial Obst-acle	Point is on Ground	Req Vert Clear 150kV (m)	Req Horiz Clear 150kV (m)
1	1	Unbuilt/agricultural area			No	No	Yes	8.5	0
2	2	Industry ground			No	No	Yes	8.9	0
3	3	Roads			No	No	Yes	8.5	0
4	4	Motorways and Highways			No	No	Yes	10.9	0
5	5	Street Lighting poles, flagpoles, sign			No	Yes	No	3.9	0
6	6	Trees which can be climbed			Yes	No	Yes	4.4	4.4
7	7	Trees which can not be climbed			Yes	Yes	No	1.9	1.9

Fig. 1. Feature code data example for an overhead power line

information system, for the coordination of all documentation related to the design phase and for the construction and execution of the system. These features help to ensure coordination between the various figures involved during the building process [4]. A model designed according to BIM criteria allows one to collect different kinds of data in a single database: terrain survey data, technical parameters and drawings (AutoCAD and PLS-CADD).

Environmental information is also included in the properties of the BIM objects, which means that the environmental properties are on the same level as other properties regarding the construction elements and may be used as a decision making criterion [5].

A. Overhead Power Line Design

The challenge in this specific field of overhead power line design was finding a way to merge the technical parameters, weather criteria, the terrain survey data and the modelling of the overhead power line into one solution. This was possible with the help of PLS-CADD software that, at the same time, allows the design team to develop a four-steps optimization process, applicable for both new overhead power lines and also for existing overhead power lines that are subject to upgrade. The four steps are described below:

- gathering terrain survey data;
- highlighting zones of prohibitive (high) cost that could be avoided;
- showing the usage of the structures;
- defining weather criteria zones along the route (weather information can be stored in different packages depending on specific weather conditions).

The optimal geographical positioning of each tower is defined along the overhead power line route with the help of the first two steps of the optimization design; the terrain survey data is helpful to determine the obstacles in terms of tower placement and conductor’s horizontal and vertical clearances. PLS-CADD has a specific tool that allows categorizing these obstacles based on type and geographic coordinates, the **Feature Code Data**, presented in Fig. 1. All fields of the Feature Code Data are configurable and adaptable to project requests and standards being applied.

The parameters of the second step can be defined as input data in PLS-CADD, thus the program will automatically avoid those areas (prohibited or at higher costs) as part of the optimization process.

Weather criteria can be defined into zones of specific climate, as the weather conditions vary depending on the length of the power line route, from high winds, low temperatures to less winds, warmer temperatures areas or icy areas [6]. An example of weather conditions used in PLS-CADD model is shown in table I.

Other factors that should be considered within the scope of this case study and verified if applicable are listed below:

- check if reusing existing structures and hardware is a feasible option;
- check if the quality of steel (steel grades, brittleness of steel) is in appropriate shape;
- verify internal and external clearances;
- verify creep and stretch conditions of conductors.

The weather criteria specifies the environmental conditions in which the line behavior is analyzed and assessed. Moreover, the weather parameters were defined and used by the application to calculate the wind pressure and loads for each wire at the attachment point on the structure.

TABLE I. WEATHER CRITERIA

Description	Air density factor (Q) (kg/m^3)	Wind speed (m/s)	Wind pressure (Pa)	Wire temperature (ºC)
Wind/10C	0.625	35,8	802	10
Extreme cold, -20C	0.625		401	-20
Maximum Operation temperature, +80C	0.625		0	80

The next step in the overhead power line design, after defining all criteria necessary for the placement of the tower and clearances for conductors, is to model the towers. PLS-CADD provides several modelling methods, of which the most complex one considers all aspects of the structure design and involves the use of TOWER software.

	Member Label	Group Label	Section Label	Symmetry Code	Origin Joint	End Joint	Ecc. Code	Rest. Code	Ratio RLX	Ratio RLY	Ratio RLZ	Bolt Type	# Bolts	# Bolt Holes	# Shear Planes	Connect Leg	Short Edge Dist. (mm)	Long Edge Dist. (mm)	End Dist. (mm)	Bolt Spacing (mm)	Shear Path Length (mm)	Tension Path Length (mm)	Rest. Coef.
1	Fg1218	L200X24	S1	XY-Symmetry	J1PF7S	J2P	1	4	1	1	1	24(8.8)	18	0	1	Both	55	125	55	120	0	0	0
2	g667	L110X12	T3	Y-Symmetry	0J4PF0.13S	0J4PF0.40S	1	4	1	1	1		0	0	0		0	0	0	0	0	0	0
3	g666	L110X12	T3	Y-Symmetry	J4PF0.40S	J4PF0.13S	1	4	1	1	1		0	0	0		0	0	0	0	0	0	0
4	g665	L110X12	T3	Y-Symmetry	0J4PF0.69S	J4PF0.40S	1	4	1	1	1		0	0	0		0	0	0	0	0	0	0
5	g663	L140x15	[illegible]	XY-Symmetry	J4PF38Y	J4PF40Y	1	4	0.5	0.5	0.5		0	0	0		0	0	0	0	0	0	0
6	g662	L140x15	S10	XY-Symmetry	J4PF43Y	J4PF46Y	1	4	0.5	0.5	0.5	24(8.8)	4	0	1	Both	55	80	55	75	0	0	0
7	g661	L200X24	S3	XY-Symmetry	J2PF13S	J2PF19S	1	4	0.5	0.5	0.5		0	0	0		0	0	0	0	0	0	0
8	g619	L200X18	[illegible]	XY-Symmetry	J3PF25X	J3PF30X	1	4	0.5	0.5	0.5		0	0	0		0	0	0	0	0	0	0
9	g664	L200X18	[illegible]	XY-Symmetry	J3PF30XY	J4XY	1	4	0.5	0.5	0.5	24(8.8)	10	0	1	Both	55	125	55	130	0	0	0
10	g763	L80X8	S12	XY-Symmetry	J4PF49S	J4PF52S	1	4	0.5	0.5	0.5		0	0	0		0	0	0	0	0	0	0

Fig. 2. Tower structure components

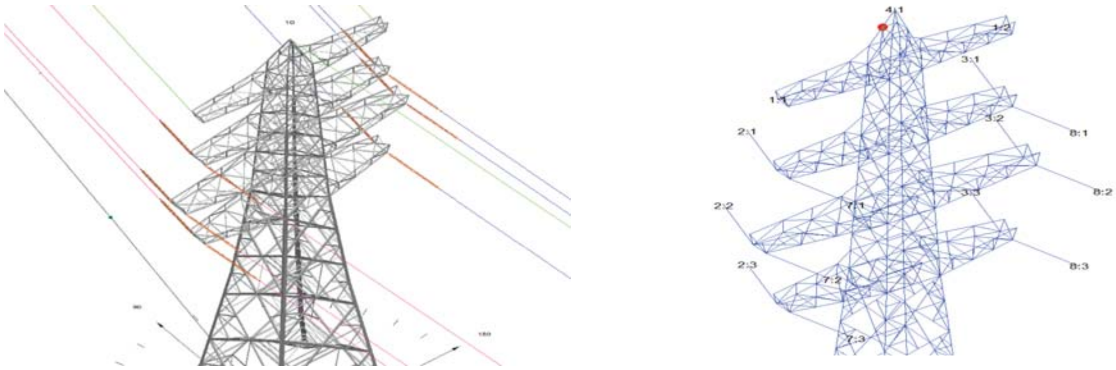

Fig. 3. Termination tower insulators arrangement - clamp insulators (1:1, 1:2. 4:1); strain insulators (2:1 – 2:3; 3:1-3:3; 7:1-7:3; 8:1-8:3)

Modelling in TOWER ensures a high level of detail including structure type, height, number of traverses, insulator placement, tensions to withstand, structure load and resulting with a 3D model of the tower.

The data in Fig. 2 shows the complexity of defining one structure's geometry in terms of joints, members, symmetry and the interconnection between them. All structure's parts are important in the pursuit of an accurate TOWER model. Besides geometry, to model the tower, complex engineering calculation was involved (wind loads, weight span and tower withstand tension after the cable stringing process). Nevertheless, the definition of traverses is essential in the linking process to PLS-CADD. The number of traversers, their lengths, the distance between them and the placement of the insulators are the parameters required in both TOWER and PLS-CADD software. Fig. 3 shows the traverses of a tower modeled in TOWER together with insulators placement (right figure) and the link into PLS-CADD with the conductors (left figure).

Merging PLS-CADD and TOWER model

To achieve an accurate design of the overhead power line, the 3D model of the termination tower designed in TOWER was integrated with the 3D line modeled in PLS-CADD.

For this to be accomplished, the insulators phase numbering and their distribution on the structures traverses, for each tower along the overhead power line route, were defined, as part of the cable stringing. The insulators used for this model were defined in three categories:

- clamp insulator for the shield wire;
- suspension insulators for suspension towers;
- strain insulators for anchor or termination towers.

Defining insulators requires the following type of parameters: insulator weight, insulator length, horizontal and vertical attachment distances to the structure. These data are important to ensure a proper design of the tower, to verify that it can withstand the tension driven from the insulators and the conductors they sustain.

Once the insulators definition and the link file are done, the next step consists of inserting the TOWER file in the PLS-CADD line model, as shown in Fig. 4. The results of merging PLS-CADD and TOWER are shown in chapter III – Data Validation.

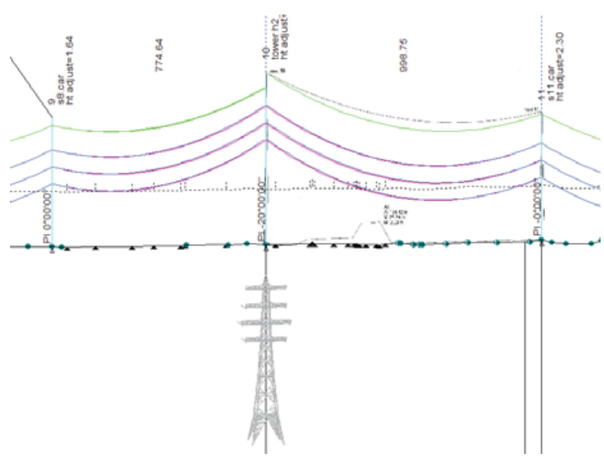

Fig. 4. Profile view of the line model with TOWER file linked

B. Substation Design

The use of BIM has the potential to improve the overall information flow, thereby achieving better project performance and quality. Moreover, the use of this tool is crucial in the project success as it enhances transparency, a better communication between the team members and a better overview of the project as a whole [7], [8].

One of the substation design phases in Revit (one of the BIM tools) includes the setup of a database of equipment families that is adaptable and reusable into other projects. The tool allows creation of parametric families which avoids reentering the same data again.

Another important phase of the substation design involves an environmental factor, delimitation of the land (the substation will occupy) and can be divided into the following steps:

- geographical coordination of the substation;
- access routes and fences;
- safety zones.

After these two phases are completed, the focus is on building up the substation. At this stage, the main difference in approach is emphasized by the level of detail required.

The base for starting the Revit model is the single line diagram of the substation. All equipment, interconnections and bays must be displayed correctly. With this modern technology, templates can be used for positioning the equipment based on safety distance and interconnection standards. Such templates (in sections and grids format) can be developed for each type of bay (example: line and cable bays, transformer bay and coupling bay). Fig. 5 shows the template for a coupling bay together with functionality correspondence in table II.

The equipment are placed at the intersection of the grids, as shown in the first column of table II. The position may vary from one bay to another depending on bay type and functionality required. The template grids VG an R show the physical boundaries of the bay and also the place where lightning poles are positioned.

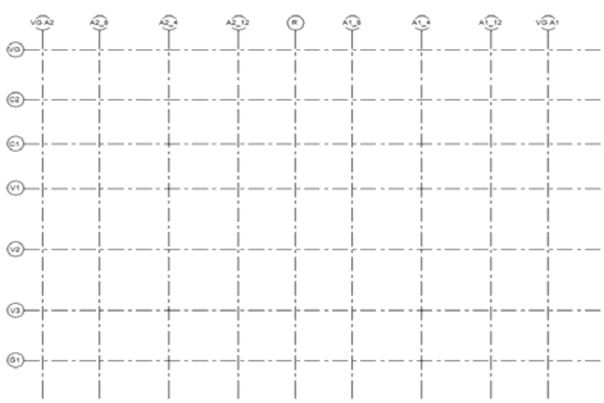

Fig. 5. Revit template for a coupling bay

TABLE II. REVIT TEMPLATE DESCRIPTION

Position in template	**Primary components**
	Description
C2	Busbar earthing switch
C1	Voltage transformer busbar
(V1, A2_12)	One pole pantograph disconnector
(V1, A1_12)	One pole pantograph disconnector and earthing switch
G1	Lightning protection poles
VG, R	Central and boundary grid lines

III. DATA VALIDATION

As an engineering design team the goal is to provide clients with not just technical results (equipment parameters and safety zones in case of short circuits), which of course are the key in providing a safe and functional substation, but also a realistic image of the overhead power line or the substation itself, the environmental impact it will have on the landscape, the area it is going to occupy and/or the length of its route (considering the terrain profile and surroundings).

These results were possible with the use of BIM environment (Revit software) and 3D modelling software like PLS-CADD and TOWER. The case study shown in this paper was conducted for a section of an overhead power line with 20 towers. This chapter emphasize the outcome of modelling the section of the overhead power line and the corresponding substation together. By conducting this case study, a complete report on structure and section usage, wind and span weight, clearances and the appropriate conductors to be installed was created.

A. PLS-CADD and TOWER Outcome

The complete overhead power line model is the result of two modelling phases: a line model along the route with structure, turning and termination tower.

Merging the two models created in TOWER and PLS-CADD in order to display the correct conductor's configuration is possible by creating a link file based on insulators orientation and numbering. The linking table thus created, takes into account not only the tower design to be integrated, but also the configuration of the adjacent towers in the back and front.

The linking parameters are shown in table III and highlights the most complex situation encountered as it gathers the following challenges:

- the tower modeled in TOWER is an anchor tower;
- the phase numbering differs between the adjacent towers at the back and the front;
- the shield wire changes configuration from three shields wires, connected two on traverse's ends and one on top of the structure, to only one shield wire on top of the structure.

The conductors (double circuit) and shield wire used to model the line are of ACSR type that operates at 150 kV. The overhead power line modelled as part of the case study is shown in Fig. 6.

TABLE III. LINKING TABLE BETWEEN TOWER AND PLS-CADD

Insulator label	Insulator type	Set number	Phase number	Set description	Termination tower
1	Strain	2	1	Phase 2-1	Yes
2	Strain	3	2	Phase 3-2	Yes
3	Clamp	4	1	Shield wire	Yes
4	Strain	7	3	Phase 7-3	Yes
5	Strain	8	2	Phase 8-2	Yes

B. *Revit and BIM Integration*

Starting from the Revit template described in chapter II/B and by implementing the families created for each type of equipment, the team was able to reach the desired outcome in the design of the substation. This was to provide an accurate model of the substation with high level of details of the components. Fig. 7 and Fig. 8 below show different modelling stages of the substation, providing top and side views of the bays.

Fig. 7 shows the equipment connection and their position inside a cable bay, for a substation of 150 kV. A detail of the cable bay is presented in the Fig. 8, highlighting the equipment's sequence for one phase.

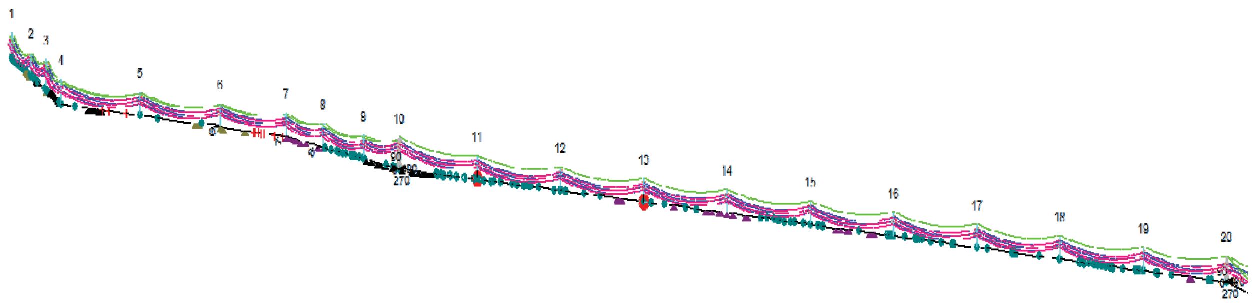

Fig. 6. PLS-CADD/TOWER model of the overhead power line – 3D view showing towers location, conductors and shield-wire representation

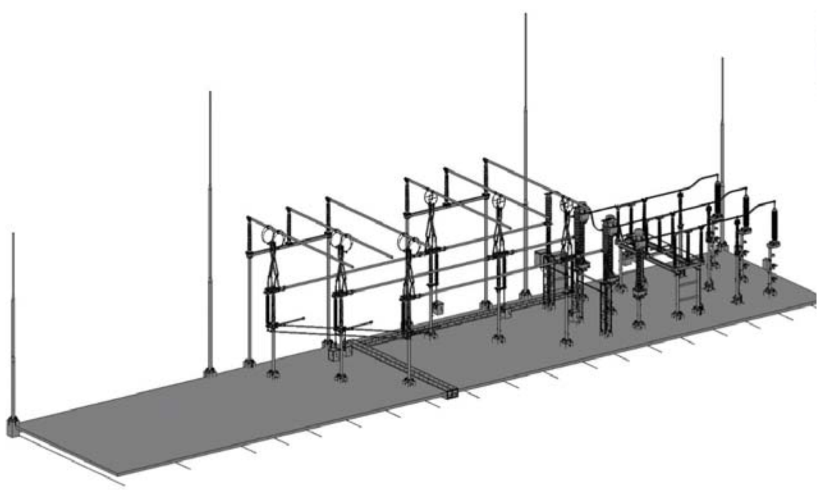

Fig. 7. Cable bay of a substation – 3D side view showing the pantograph, circuit breaker, combination transformer, earthing switch and cable terminations

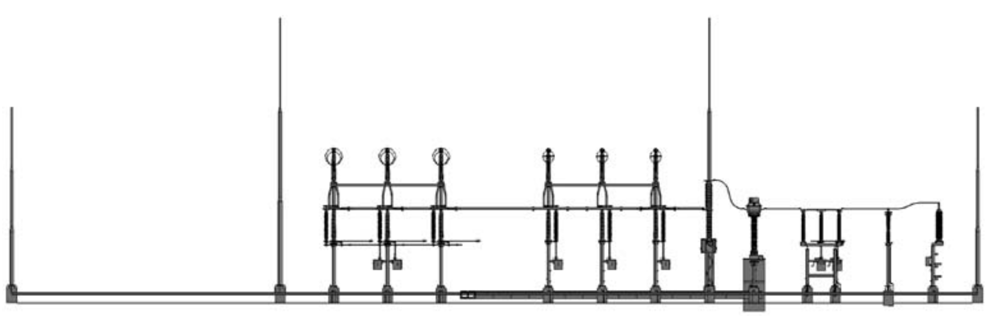

Fig. 8. Cable bay of a substation – 2D view detail

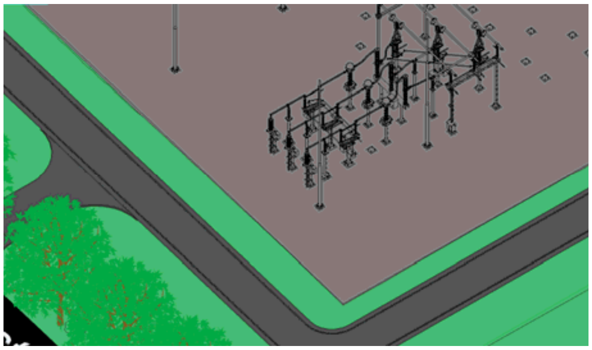

Fig. 9. Substation layout including top surface elements

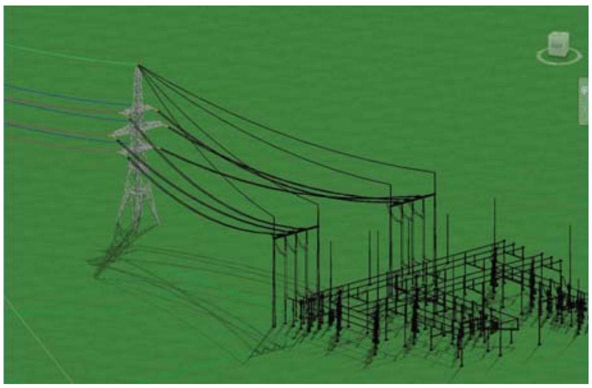

Fig. 10. Merging PLS-CADD/TOWER & Revit into one design

The quality of the model was enhanced by applying rendering effects that imitates real life situation. The terrain profile was filled in with the correspondent elements (environment: grass, water areas, trees; weather conditions: sunny, sunny with few clouds being applied too). The results obtained are presented in Fig. 9.

Once the overhead power line and the substation models were completed, another challenge was to integrate the two models in the same BIM environment and see it all as a whole. The compatibility and flexibility of PLS-CADD, TOWER and Revit software allowed this merge and made it possible to deliver a great design result in 3D (see Fig.10).

IV. Conclusion

The aim of the paper is to emphasize the importance of BIM integration in the modelling of the high voltage power systems as it is a huge step towards time saving and sustainability.

The use of these modelling tools has shown high level of adaptability in providing both simple to very complex solutions in terms of engineering analysis for the overhead power lines and substation design.

The modelling outcomes presented in this paper have shown the possibility to work with different software and to provide an unique solution by embodying the models build into the BIM environment.

In one hand, PLS-CADD and TOWER tools have led to great results in the design of overhead power line (including the 3D tower modelling); in the other hand the use of Revit had make it possible to develop 3D models of high voltage substations with details on the bays, equipment and surroundings.

An important aspect lays, likewise, in the compatibility between the two pieces of software which makes possible combining a solution with both technical and high visual impact. The automatization process integrated in these software, as presented in this case study, is what makes it a time saving choice and cost effective for modelling high voltage power systems. With this process a design framework (including component types, cable types, different criteria, substation bay layouts) can be created, reused and adapted to customer requirements.

The BIM environment is the right step in the evolution of high voltage power systems design as it gives high flexibility in terms of software use, versatility and implementation of design frameworks that are reusable and adjustable to new requirements.

Acknowledgment

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